

SM fibre is more expensive due to the small central core of the fibre-optic cable. It requires a bright and powerful light source to use the cable.

SM-distributed feedback lasers are inherently more expensive than MM vertical-cavity-surface-emitting lasers (VCSEL), and this cost differential continues indefinitely. The higher cost of SM transceivers translates to a higher total channel cost in spite of the fact that SM fibre is cheaper than MM fibre. But VCSEL sources require many more fibres, and that can offset the higher cost of SM wavelength division multiplexing (WDM) transceivers. SM fibre at 100G is especially advanced, using 4x25G channels over each fibre, making it much less complex in the cable plant.

Due to its unlimited bandwidth and distance, SM fibre is becoming more popular in data centres. Connections to the Internet coming in from the outside world will be on SM fibre, as it allows more flexibility in the number of interconnections. SM transceivers are more expensive but cabling is much cheaper and easier to install, upgrade and manage. SM fibre has advantages of testing time and cost, probably enough to balance out the higher cost of transceivers.

Optimised fibre-optics solutions

Choice of fibre usually depends on data centre design choices. Laser-optimised 50/125µm OM3/OM4 MM fibre channel is the primary method to reliably link servers to external data storage devices in enterprise data centres. Bend-insensitive fibre performs better in cramped cable trays

TABLE III MULTI-MODE FIBRE VERSUS SINGLE-MODE FIBRE		
Cabling	40G	100G
MM parallel	Eight fibres, four Tr - four Rcv, four unused	20 fibres, 10Tr - 10 Rcv, four unused
MM WDM (OM5) SM WDM	Two fibres	Two fibres

and racks. Even inside the servers, increasing speed and reducing power has led to the need for optical fibre interconnect development.

The idea developed by Cisco and Finisar is making MM fibre a possibility for up to 400G, or at least 100G on only two fibres. It uses 840nm - 940nm wavelength range with 30nm spacing WDM, with VCSELs on an upgraded OM4 fibre, now standardised as OM5. Cisco is already using this technology and, while it has been standardised, it is not widely available commercially.

Optical-fibre cabling will be the only network connectivity capable of supporting data rates of 50Gbps and beyond in typical network applications. This can be demonstrated by referencing state-of-the-art 50Gbps network switches. These new high-speed switches typically contain a 50Gbps application specific integrated circuit (ASIC) designed to support 256Gbps - 50Gbps switch ports. For these large port counts, it is advantageous to aggregate 8Gbps - 50Gbps ASIC I/O ports to produce 1Gbps - 400Gbps optical switch ports. Aggregation reduces the number of switch ports from 256Gbps - 50Gbps ports to 32Gbps - 400Gbps ports in a 1RU form factor. These high-speed high-port-density network switches have signifi-

cant consequences on network connectivity.

As the need for bandwidth increased, the industry began moving towards networks and cabling with higher fibre densities. Because of the increase in the number of fibres

used with parallel optics, an efficient, high-density interconnect was needed.

To accommodate the high volume of connectors within ever-tighter space constraints, installers and designers created a multi-fibre push-on (MPO) connector. The MPO format dramatically reduced the amount of time, effort and space required to install and deploy network technologies, particularly in parallel optic applications. Highly-adaptable and flexible, MPO connectors have continued to evolve to meet the emerging needs of installers, data centres and organisations that rely on them.

Future options

Unlike the parallel 10G channels used with MM fibre, SM uses WDM to transmit multiple channels over one fibre at different wavelengths. Thus, 40G uses four 10G wavelengths and 100G is achieved using four 25G wavelengths.

Parallel optics is the preferred technology due to cost and safety factors. Although the cost of cabling is lower for SM WDM, cost benefits are not realised until the channel reach exceeds 100m, which exceeds the maximum reach for the majority of MM fibre applications. Transceivers emit high optical power levels to compensate for high insertion loss of WDM devices, which approach eye safety limits.

Ongoing development for high-performance servers and storage technologies drives the need for increased fibre channel data rates to reliably link these devices. This will maximise operating efficiencies and enable low-cost value propositions. The fibre channel's deterministic data delivery, low latency and proven reliability have made it the leading transport technology for linking servers to external data storage. **EFY**

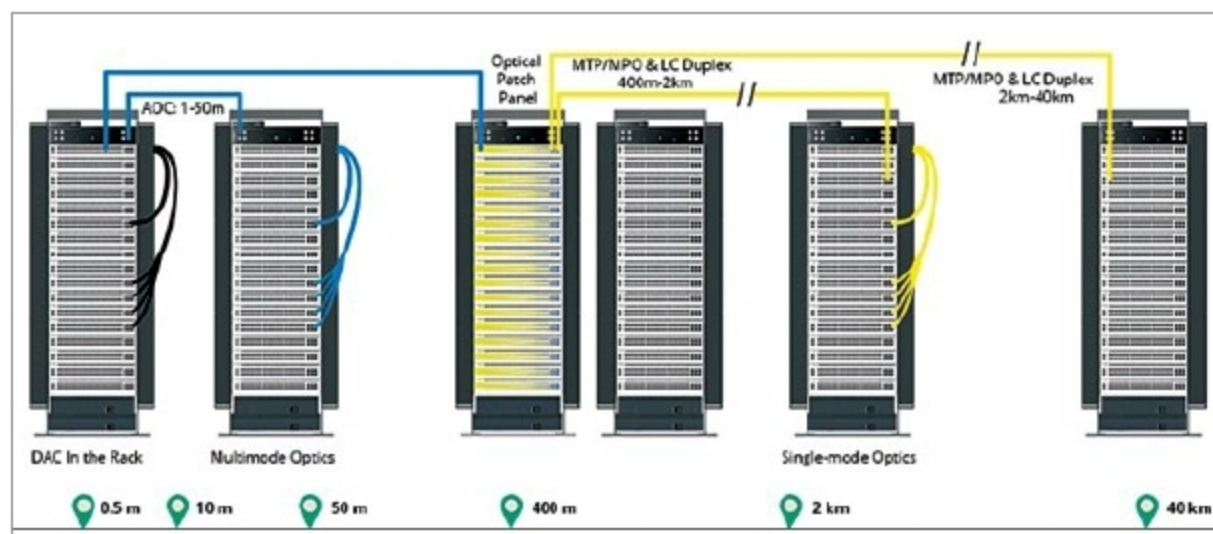


Fig. 2: 40G cabling solution for data centre (Credit: www.fs.com)